Project Initialization and Planning Phase

Date	28 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Define Problem Statements:

Farmers often face difficulties in selecting the most suitable crop due to a lack of knowledge about soil nutrients and changing environmental conditions. Incorrect crop selection leads to low productivity, resource wastage, and financial losses. Therefore, there is a need for an intelligent machine learning-based system that analyzes soil and environmental parameters such as Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall to recommend the most suitable crop and support sustainable agricultural decision-making.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A farmer seeking suitable crops for cultivation	Select the best crop based on soil and environmental conditions	I don't know which crop is suitable for my land and weather conditions	I lack proper knowledge about soil nutrients, rainfall, and climate conditions	Worried about crop failure and financial losses
PS-2	An agricultural researcher	Analyze agricultural data to improve farming decisions	Processing large amounts of soil and environmental data manually is difficult	Traditional methods are timeconsuming and less accurate	Frustrated and limited in making datadriven decisions

PS-3	A small-scale farmer	Increase crop productivity and profit	I often choose crops based on guesswork	I do not have access to intelligent farming guidance	Uncertain about my farming future